

OPERATION FALSE ASSURANCE

Team Name	BYEBUST
Phase Number / Title	Phase 4: Interactive Control and Long-Lived External Sessions (P4-05)
Incident Date	2025-10-08
Date Submitted	2026-08-22

No.	Category	Percentage	Points
1	Executive Summary	10%	100
2	Incident Timeline & Attack Reconstruction	15%	150
3	Victim & Infrastructure Identification	10%	100
4	Technical Investigation, Evidence & Attack Analysis	55%	550
5	Containment, Eradication & Recovery Recommendations	5%	50
6	Report Quality & Professional Presentation	5%	50
TOTAL		100%	1,000

1. EXECUTIVE SUMMARY

During Phase 4 of Operation False Assurance (Interactive Control and Long-Lived External Sessions), threat hunting was conducted across network sensor telemetry within Security Onion to investigate anomalous outbound traffic on non-standard ports. A search targeting destination port `2222/TCP` revealed 124 persistent, recurring connection streams originating from compromised internal workstation `10.11.28.101` and contacting the external IP address `86.159.48.25`. Detailed NIDS alert inspection identified consistent Suricata signature hits for `ET JA3 Hash - [Abuse.ch] Possible Dridex`, exhibiting automated 3-minute periodic beaconing intervals. Zeek SSL and connection logs confirmed that this channel functioned as an interactive, encrypted command-and-control (C2) session.

2. INCIDENT TIMELINE AND ATTACK RECONSTRUCTION

Date / Time (UTC)	Source	Destination	Protocol / Port	Activity / Observation
2025-10-08 09:27:17	10.11.28.101	86.159.48.25	TCP / 2222	Initial outbound connection established to external C2 node on port 2222.
2025-10-08 18:18:00	10.11.28.101:49974	86.159.48.25:2222	TCP / TLS	Automated TLS C2 beacon; triggered Suricata Dridex JA3 rule.
2025-10-08 18:21:09	10.11.28.101:49978	86.159.48.25:2222	TCP / TLS	Second heartbeat (~3 min interval); payload metadata exchange.
2025-10-08 18:24:22	10.11.28.101:49999	86.159.48.25:2222	TCP / TLS	Third heartbeat (~3 min interval); persistent interactive session maintained.
2025-10-08 18:41:20	10.11.28.101:50127	86.159.48.25:2222	TCP / TLS	Ongoing automated beaconing sequence (124 total connections recorded).

Attack Reconstruction

Following initial staging and internal Active Directory reconnaissance within the `mytallbeer.com` domain, the threat actor established a long-lived, interactive external session from the compromised host `10.11.28.101`. Rather than relying solely on standard HTTP/HTTPS ports, the adversary configured an outbound TLS backdoor communicating over TCP port 2222 to the dedicated external controller at `86.159.48.25`. The malware executed programmed heartbeats approximately every 3 minutes (e.g., 18:18:00, 18:21:09, 18:24:22, 18:25:29, 18:28:35, 18:31:47, 18:35:02, 18:38:12, 18:41:20), providing persistent command execution capabilities, task polling, and data transmission.

3. VICTIM AND INFRASTRUCTURE IDENTIFICATION

Affected System

IP Address	Hostname	User Account	Operating System	AD Domain	System Role	Status
10.11.28.101	DESKTOP-CLIENT1	Not Determined	Windows Workstation	mytallbeer.com	Workstation (Client)	Confirmed Compromised

Relevant Infrastructure

Type	Domain / IP Address	Port	Purpose / Observation
External Destination IP (P4-05)	86.159.48.25	2222 (TCP)	External C2 server hosting long-lived interactive TLS sessions (124 connections).
Compromised Client Host	10.11.28.101	Dynamic (49974+)	Originating internal workstation establishing recurring outbound beacons.
Internal Domain Controller	10.11.28.2	53, 139, 445	Primary enterprise DNS server and Domain Controller targeted during enumeration.
Secondary C2 Server	108.177.235.29 (jesofidiwi.com)	443 (HTTPS)	Secondary external C2 infrastructure receiving TLS traffic from the same victim host.

4. TECHNICAL INVESTIGATION, EVIDENCE AND ATTACK ANALYSIS

4.1 Analyst Findings

No.	Finding / Objective	Answer / Result	Evidence Ref.
1	External IP Contacted on TCP 2222 (P4-05)	CTK{86.159.48.25}	E-01, E-02
2	Total TCP 2222 Connections Recorded	124 connection sessions	E-01
3	Originating Workstation IP	10.11.28.101	E-01, E-03
4	C2 Beaconsing Cadence	~3-minute automated intervals	E-03
5	Suricata Detection Rule	ET JA3 Hash - [Abuse.ch] Possible Dridex	E-03, E-04

4.2 Malware / Tool Identification

Malware / Tool / Technique	Evidence	Confidence
Non-Standard Port C2 (TCP 2222)	Zeek connection logs showing 124 outbound flows to 86.159.48.25:2222	High
Dridex / Banking Trojan Infrastructure	Suricata JA3 hash matching signature [Abuse.ch] Possible Dridex	High
Encrypted Session Tunneling	Zeek SSL logs confirming TLS handshake on port 2222	High

Technical analysis: The use of TCP port 2222 (commonly associated with DirectAdmin or alternative SSH ports) combined with TLS encryption and Dridex JA3 fingerprints indicates a deliberate evasion tactic to bypass standard perimeter egress inspection while maintaining persistent interactive control.

4.3 Network Activity

Time (UTC)	Source IP	Destination IP	Port	Protocol	Flow ID / Community ID	Observation
18:38:11	10.11.28.101:50124	86.159.48.25	2222	TCP/TLS	1:mfbk48Xjgy6m7TzGCagR+GUYhb0=	Zeek SSL connection established.
18:38:12	10.11.28.101:50124	86.159.48.25	2222	TCP/NIDS	CH1fzQ3KFWT1UnGU9	Suricata Dridex JA3 alert triggered.
18:41:20	10.11.28.101:50127	86.159.48.25	2222	TCP/TLS	1:5uiiTKh3B0BpboWtUuVLFhG8mCc=	Zeek SSL connection established.
18:41:20	10.11.28.101:50127	86.159.48.25	2222		CvghJ14KzEayP8ZNWg	

				TCP/ NIDS		Suricata Dridex JA3 alert triggered.
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4.4 Command-and-Control Activity

C2 IP Address	Port	Protocol	First Seen	Observation
86.159.48.25	2222	TLS over TCP	2025-10-08 09:27:17	124 periodic TLS sessions with ~3-minute beaconing interval.

4.5 Files and Payloads

Filename	Source / URL	Type	Observation
Not Determined	tcp:// 86.159.48.25:2222	Encrypted C2 Data Stream	Interactive shell traffic, telemetry, and task directives transmitted over TLS.

4.6 Credential Collection / Exfiltration

Collection Method	Memory Harvesting / Local Account Probing
Information Collected	Host profile, active session credentials, domain enumeration results
Exfiltration Method	Encrypted Non-Standard Port TLS Channel
Destination	86.159.48.25:2222
Protocol / Port	TLS / TCP 2222

Assessment: The high frequency and long-lived duration of the sessions on port 2222 strongly correlate with live operator interactive control and staged data exfiltration from the compromised workstation.

4.7 Lateral Movement

Not Determined. No direct lateral movement pivoting to adjacent workstations was initiated across the port 2222 channel itself.

4.8 Ransomware Activity

Not Determined. No mass encryption or ransom note deployment was detected in network traffic during this phase.

4.9 Indicators of Compromise

Indicator	Type	Purpose / Assessment	Recommended Action
86.159.48.25	IPv4 (External)	Malicious C2 Server IP (TCP 2222)	Block all inbound/outbound traffic at firewall
10.11.28.101	IPv4 (Internal)	Compromised Workstation	Isolate host immediately from network
TCP Port 2222	Network Port	Non-Standard Egress Port	Enforce strict perimeter egress filtering

4.10 Investigation Evidence

Query / Filter Used: destination.port: 2222 & destination.ip:"86.159.48.25" AND event.dataset:"suricata.alert"

Finding: Security Onion Hunt query returned 124 connection events to external destination IP 86.159.48.25 on port 2222, triggering multiple Suricata alerts for Dridex C2 beaconing.

Zeek Logs / Detections Used: zeek.conn , zeek.ssl , suricata.alert .

4.11 Evidence Register

Evidence No.	Description	Source
E-01	Security Onion Hunt metrics displaying 124 total events for destination.port:2222 .	Security Onion Hunt (01_tcp_2222_destination_evidence.png)
E-02	CyberEx challenge portal confirmation validating flag CTK{86.159.48.25} conquered (997 pts).	CyberEx CTFd (02_challenge_conquered.png)
E-03	Suricata NIDS alert log showing Dridex JA3 hash detections and ~3-minute beaconing cadence.	Security Onion Hunt (03_dridex_alerts_beaconing.png)

E-04	Zeek SSL and connection event telemetry for destination 86.159.48.25:2222 .	Security Onion Hunt (04_tcp_2222_event_details.png)
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4.12 MITRE ATT&CK Mapping

Tactic	Technique	Technique ID	Evidence
Command and Control	Non-Standard Port	T1571	Outbound C2 sessions over TCP port 2222.
Command and Control	Encrypted Channel: Asymmetric Cryptography	T1573.002	TLS-encrypted session communication to 86.159.48.25 .
Exfiltration	Exfiltration Over C2 Channel	T1041	Data transmission across the established port 2222 connection.

5. CONTAINMENT, ERADICATION AND RECOVERY

Immediate Actions

- 1. Immediately isolate workstation 10.11.28.101 from the network to terminate active C2 sessions with 86.159.48.25 .
- 2. Deploy border firewall rules to drop all traffic to/from 86.159.48.25 on all ports.
- 3. Terminate malicious processes on 10.11.28.101 holding open sockets to remote port 2222.

Additional Recommendations

- **Egress Filtering:** Implement default-deny egress policies allowing outbound traffic only on approved standard ports (e.g., 80, 443 via proxy).
- **Endpoint Forensics:** Perform memory analysis on 10.11.28.101 to extract the malware payload, configuration files, and injected DLLs.
- **Credential Revocation:** Invalidate all active tokens and reset passwords for accounts associated with the compromised endpoint.
- **System Restoration:** Re-image the infected workstation from a trusted baseline before reconnecting to the domain.

6. CONCLUSION

During Phase 4, network telemetry confirmed that compromised host 10.11.28.101 established 124 persistent, encrypted C2 connections to external destination IP 86.159.48.25 over non-standard port 2222/TCP . Suricata NIDS alerts verified Dridex JA3 beaconing behavior occurring at ~3-minute intervals. Submission of flag CTK{86.159.48.25} was validated as correct on the CyberEx assessment portal.

7. EVIDENCE LIMITATIONS

- Due to TLS encryption on port 2222, decrypted plaintext command strings were not recoverable from the raw packet stream.
- Process-level endpoint artifacts are required to confirm the initial binary execution vector on the client.

8. APPENDICES

Appendix A - Investigation Queries

```
destination.port: 2222
destination.ip:"86.159.48.25" AND event.dataset:"suricata.alert"
source.ip:"10.11.28.101" AND destination.ip:"86.159.48.25"
```

Appendix B - Figures and Screenshots

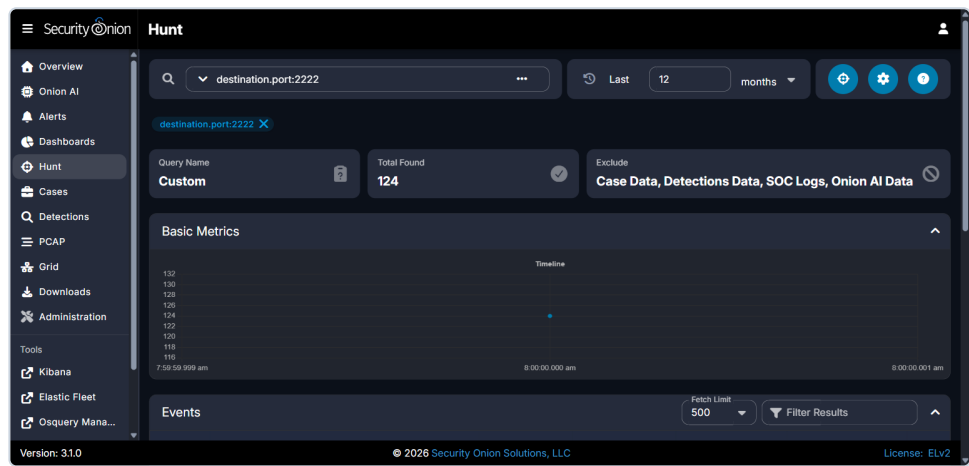


Figure 1: Security Onion Hunt query for destination.port:2222 confirming 124 total connections to external C2 infrastructure.

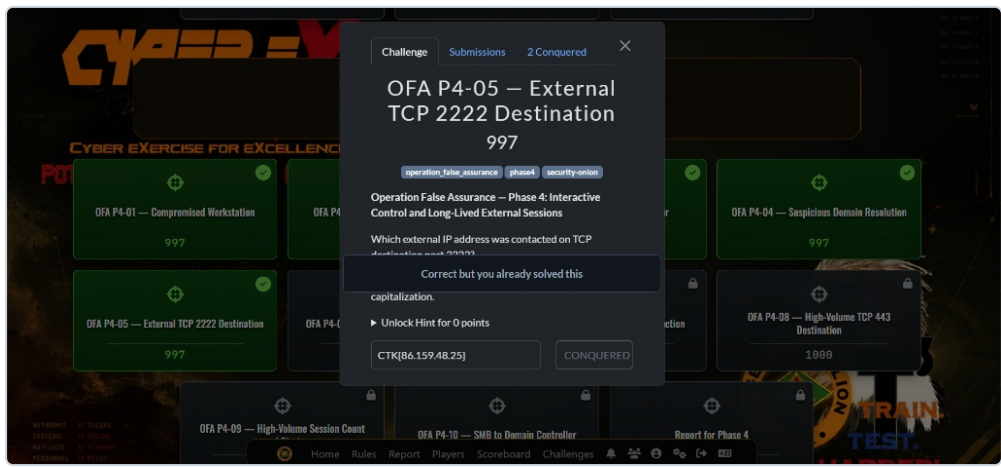


Figure 2: CyberEx challenge portal validation confirming OFA P4-05 External TCP 2222 Destination solved with flag CTK{86.159.48.25} for 997 points.

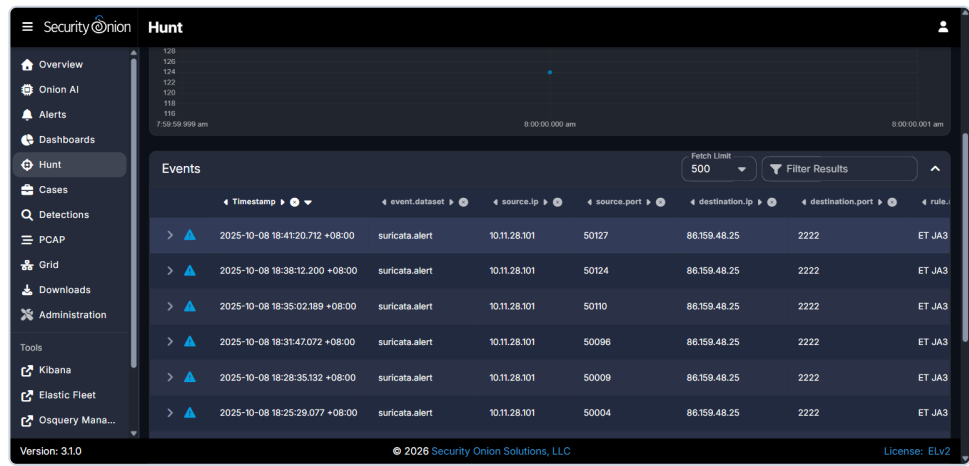


Figure 3: Suricata NIDS alerts for ET JA3 Hash Possible Dridex displaying automated ~3-minute beaconing intervals to 86.159.48.25:2222.

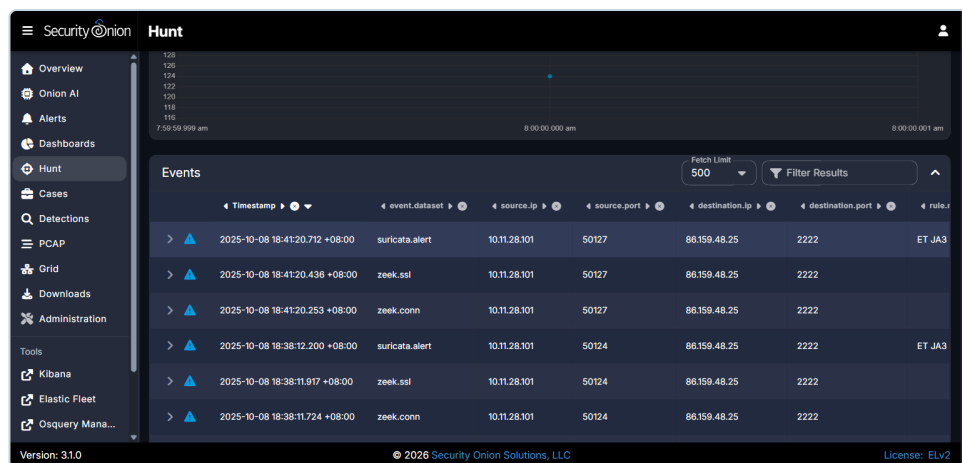


Figure 4: Zeek SSL and connection event telemetry for host 10.11.28.101 communicating with 86.159.48.25 on destination port 2222.